

Acoustic Emission and Passive Acoustic Regression to the ISO 281 Viscosity Ratio κ for Online, Non-Invasive Lubrication Condition Monitoring of Ball Bearings: A Controlled Dual-Channel Study

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Abstract

Inadequate lubrication is responsible for an estimated 30–80% of rolling element bearing failures, yet reliable online, non-invasive lubrication condition monitoring (LCM) remains an unsolved engineering problem. Passive sensor-based approaches are most practical for industrial retrofit because they require no bearing modification, but the signal processing challenge of separating lubrication-relevant information from confounded operating-condition effects has not been systematically characterised for simultaneous acoustic emission (AE) and passive ultrasound (US) sensing. This paper presents a controlled study in which AE and passive US signals are simultaneously acquired from a ball bearing during staircase speed sweeps (140–2977 rpm) at temperatures spanning 33–100 °C, covering the boundary, mixed, and full-film lubrication regimes across 8 504 sweeps per channel. The ISO 281 viscosity ratio $\kappa \in [0.04, 1.64]$ (mean 0.51) is derived analytically from bearing geometry, lubricant properties, and operating conditions as a continuous, physics-grounded regression target. For the AE channel, 14 canonical time-domain and frequency-domain features are extracted from the broadband signal and from three frequency sub-bands (20–500 kHz, 500 kHz–1 MHz, 1–2 MHz), yielding 56 candidate features; 42 candidate features are extracted from the US channel (broadband 0–20 kHz and two sub-bands: 0–10 kHz and 10–20 kHz). A two-stage selection retains 27 AE and 10 US features after a joint Spearman–Pearson correlation threshold filter and a greedy variance-inflation-factor and pairwise-correlation redundancy reduction. Three regression model families are benchmarked across three sensor configurations using repeated nested 5-fold cross-validation ($R = 5$ repetitions, 25 outer scores per model); LightGBM on AE features alone achieves a hold-out performance of $R^2 = 0.953$, RMSE = 0.079 κ -units. Fusing AE and US features yields a small but consistently significant improvement of $\Delta\text{RMSE} = 0.0079$ (Nadeau–Bengio corrected t -test $p = 1.5 \times 10^{-5}$; hold-out Wilcoxon $p = 5.5 \times 10^{-16}$, Diebold–Mariano $p = 1.6 \times 10^{-8}$, bootstrap 95% CI [0.0053, 0.0105] excluding zero), indicating that the heterodyned US channel carries a modest amount of non-redundant κ information. Significance of pairwise model differences is assessed by Nadeau–Bengio corrected repeated k -fold t -tests with Holm–Bonferroni correction for within-sensor comparisons, and by Wilcoxon signed-rank, Diebold–Mariano, and bootstrap ΔRMSE confidence intervals for cross-sensor comparisons.

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1 Introduction

Rolling element bearings are among the most prevalent components in industrial machinery, and their reliable operation depends critically on adequate lubrication. Lubrication failure—through starvation, viscosity degradation, or contamination—increases metal-to-metal asperity contact, accelerates surface fatigue and wear, raises operating temperature, and ultimately causes premature failure. Improper lubrication is estimated to account for between 30% and 80% of all rolling element bearing failures [1–3].

Despite this, lubrication condition monitoring (LCM) has received far less attention than vibration-based fault detection and remaining-useful-life estimation, which are now mature and widely deployed [4, 5]. The lubrication signal is harder to exploit: its acoustic and vibratory signatures are weaker than the impulsive signatures of spalling or raceway cracks and are strongly entangled with speed and load, so operating-condition effects readily mask the lubrication-dependent component [5–7]. Yet accurate online LCM would enable condition-based relubrication—applying lubricant when and in the quantity actually needed, rather than on fixed schedules that systematically under- and over-lubricate [8].

The dominant bearing-monitoring approach, vibration analysis, detects macroscopic faults such as raceway spalling and rolling-element damage through characteristic-frequency analysis, but is inherently insensitive to the subtle tribological changes of film formation and breakdown [5, 9]. Temperature monitoring and oil analysis add complementary information but are indirect, subject to sampling delay, or require access to the lubricant reservoir, and none provides a direct, continuous, non-invasive indicator of the instantaneous lubrication regime under operating conditions [8].

Passive, non-invasive sensing, e.g. acoustic emission (AE) and passive ultrasound (US), is practically attractive for industrial LCM because it requires no bearing modification, no direct access to the lubricant, and can be retrofitted to bearings in service. AE captures the broadband stress waves emitted by asperity contact and microcrack events, while passive US underlies the threshold-based relubrication tools already in routine industrial use, in which the broadband ultrasonic level is compared against a baseline and relubrication is triggered once it rises by a set margin [1, 10, 11]. The physical origins of both signals are detailed in Section 2. Despite the practical maturity of threshold-based US and the demonstrated usefulness of AE, the two channels’ signal-processing behaviour has not been characterised simultaneously within a single regression framework against a physics-grounded lubrication reference.

The sensitivity of AE to lubrication regime is well established. Miettinen et al. [12] showed that AE pulse count and RMS distinguish boundary, mixed, and full-film regimes and respond to base-oil viscosity and thickener content in grease-lubricated bearings. Yoshioka and Shimizu [13] combined AE and vibration monitoring on a grease-lubricated deep-groove ball bearing using envelope demodulation and statistical AE parameters.

Tandon et al. [14] found AE and shock-pulse sensors to outperform accelerometers for detecting grease contamination in a ball bearing—the scenario closest to the present study. More recently, Cornel et al. [9] showed that AE detects lubrication-sensitive surface damage such as micro-pitting and smearing up to eight hours earlier than vibration monitoring, and Hou and Zhang [15] derived amplitude- and energy-centre features that classify lubrication status across speeds and viscosity levels, though without a continuous regression target, and with no variation of temperature.

On the signal-processing side, Renhart et al. [16] applied octave filter-bank decomposition with RMS–friction features to characterise AE tribological events, and Marticorena and García Peyrano [6] showed that spectral correlation of AE resolves lubrication-regime changes under varying load, demonstrating that kinematic and acoustic features can partially decouple operating-condition confounds. The physical link between film adequacy and the AE signature has been formalised by Liu et al. [17] and Zhang et al. [7], whose coupled vibration–AE models tie emission intensity and frequency content to EHL film thickness and the rate of asperity contact. For the passive-ultrasound channel, Jakobsen et al. [1, 11] showed that broadband acoustic features carry measurable information about the ISO 281 viscosity ratio κ .

A recurring limitation of data-driven LCM is its reliance on discrete fault labels or qualitative lubrication states rather than a physically interpretable, continuous target. The ISO 281 viscosity ratio κ removes this limitation: it is the ratio of the lubricant’s actual kinematic viscosity at operating temperature to the minimum viscosity required for full EHL film formation at the current speed and bearing geometry, mapping the instantaneous operating point to a standardised lubrication-adequacy index (Section 2). A model trained to predict κ implicitly learns the joint dependence on speed, temperature, and lubricant properties, and its output is directly interpretable against established regime boundaries. Despite this, κ has seldom been used as a regression objective in acoustic condition monitoring.

The clearest precedent is the work of Jakobsen et al. [1, 11], who established κ regression for passive bearing acoustics, predicting κ from low-cost MEMS microphone and accelerometer features using LASSO and neural networks. The present study adopts their κ target and extends it in three respects: it adds a calibrated wideband AE channel acquired simultaneously with the passive US channel, extracts frequency sub-band features from both signals, and tests whether the two modalities are complementary. This last point is an open question: AE–US complementarity is argued on physical grounds in reviews [5, 18] but has not been quantified experimentally, and it carries direct practical weight, since a second sensor adds cost and complexity and is justified only if it supplies non-redundant information. No prior study has simultaneously acquired AE and passive US from a single ball bearing and quantitatively assessed their joint value for κ regression.

This study deliberately fixes the bearing, lubricant, and load, varying only rota-

tional speed and temperature. Holding the mechanical configuration constant isolates the lubrication-dependent acoustic response from the bearing-to-bearing variability and degradation processes that would otherwise confound it—a prerequisite for establishing whether AE and passive US carry κ -relevant information at all. The resulting limits on generalisation across bearing types, lubricants, and load regimes are genuine, but not addressed here.

Contributions

This paper makes the following contributions:

1. **A dual-channel LCM dataset.** The first dataset to pair calibrated AE and passive-US signals from a single ball bearing across a continuous speed (100–3000 rpm) and temperature (40–100 °C) sweep spanning the boundary, mixed, and full-film regimes, with an analytically derived ISO 281 κ ground truth.
2. **Slipring based film thickness measurement**
3. **Features of sub-bands of AE signals predict κ .** LightGBM on AE features alone reaches hold-out $R^2 = 0.953$ (RMSE = 0.079 κ -units); SHAP attribution identifies the high-frequency 1–2 MHz AE sub-band as the single most informative range for κ prediction.
4. **AE substantially outperforms passive (heterodyned) US.** The AE–US gap ($\Delta R^2 = 0.14$) is confirmed significant across all tests, quantifying the value of wideband AE over a heterodyned passive-US channel.
5. **The AE–US fusion question is answered quantitatively.** Combining the two channels yields a small but statistically robust gain ($\Delta \text{RMSE} = 0.0079$), significant under repeated nested cross-validation ($p = 1.5 \times 10^{-5}$) and confirmed on the independent hold-out set—directly addressing the open question of whether a second acoustic modality supplies non-redundant κ information.

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The remainder of the paper is organised as follows. Section 2 reviews the theory of κ and the physical origins of AE and US signals. Section 3 describes the experimental setup and data collection. Section 4 presents the signal-processing and feature-extraction pipeline. Section 5 reports the feature analysis. Section 6 presents the regression modelling and statistical evaluation. Section 7 discusses the findings, and Section 8 concludes.

The overall pipeline of the experiment, signal processing, modeling, and evaluation is seen in Figure 1.

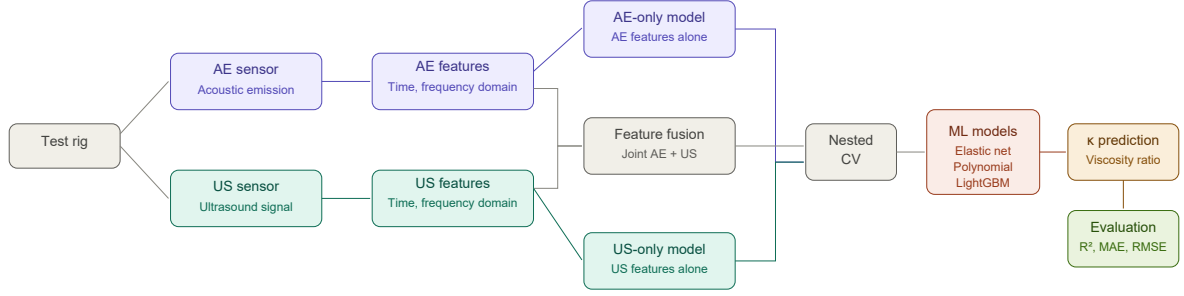


Figure 1: Overview of the experimental and modelling pipeline. AE and passive US signals are acquired simultaneously from the test rig and decomposed into broadband and sub-band features. Three sensor configurations (AE-only, AE+US fusion, US-only) are evaluated across three regression model families under repeated nested cross-validation, with the viscosity ratio κ as the regression target.

2 Background

2.1 Lubrication regimes and the viscosity ratio κ

Rolling element bearing contacts operate in one of three lubrication regimes depending on whether the lubricant film is sufficient to fully separate the rolling surfaces. In the full-film elastohydrodynamic (EHL) regime the surfaces are completely separated and asperity contact is suppressed; in the mixed regime the film is partially formed and intermittent asperity contact occurs; in the boundary regime the film is inadequate and metal-to-metal contact dominates. The ISO 281 standard characterises these regimes through the viscosity ratio κ , with boundaries at $\kappa < 0.5$ (boundary), $0.5 \leq \kappa < 1$ (mixed), and $\kappa \geq 1$ (full film).

The ISO 281 standard defines the viscosity ratio

$$\kappa = \frac{\nu}{\nu_1(n, d_{pw})}, \quad (1)$$

where ν [$\text{mm}^2 \text{s}^{-1}$] is the actual kinematic viscosity of the lubricant at operating temperature, and ν_1 is the minimum kinematic viscosity required for full EHL at the bearing's rotational speed n [rpm] and mean pitch diameter d_{pw} [mm] [19]. The reference viscosity is given by

$$\nu_1 = \begin{cases} 45\,000 \, n^{-0.83} \, d_{pw}^{-0.5} & n < 1000 \text{ rpm}, \\ 4\,500 \, n^{-0.5} \, d_{pw}^{-0.5} & n \geq 1000 \text{ rpm}. \end{cases} \quad (2)$$

The operating viscosity ν is computed from the ASTM D341 Walther viscosity–temperature equation fitted to the lubricant’s 40 °C and 100 °C reference values, with a standard value of $\lambda = 0.7$. Since ν_1 decreases with increasing rotational speed—faster rotation enhances hydrodynamic film entrainment, requiring less viscosity to achieve full EHL— κ captures the combined effect of speed, temperature, and lubricant grade on lubrication adequacy at a given operating point. Importantly, κ assumes nominally intact lubricant: the Walther equation reflects the properties of fresh grease, so κ does not encode viscosity changes arising from lubricant degradation, contamination, or starvation. Extending the regression framework to track such changes would require either continuous viscosity updates or an alternative target variable.

2.2 Physical origins of AE and passive ultrasound signals

Acoustic emission (AE) refers to transient elastic stress waves generated by sudden energy releases in a material: asperity contact and micro-fracture are the dominant sources in rolling bearings [12, 16]. Under boundary and mixed lubrication, intermittent metal-to-metal asperity contacts generate broadband stress waves that propagate through the bearing structure and are detected by piezoelectric AE sensors coupled to the outer housing [9, 16]. The emission rate, amplitude, and frequency content of these events are all sensitive to the contact severity and hence to the lubrication regime [9, 12]. Full-film EHL suppresses asperity contact, reducing AE activity; the transition through the mixed regime is associated with intermittent, structurally distinct emission events. AE sensors for bearings respond from tens of kHz into the MHz range and require correspondingly high sampling rates to capture the full frequency content.

Passive ultrasound sensing detects structure-borne acoustic energy above the audible range. As a modality it is not intrinsically narrowband—ultrasonic emissions span from roughly 20 kHz into the MHz range, overlapping much of the AE band—but the instruments most widely deployed for bearing monitoring operate in a narrow band around 38–40 kHz. In this band the sensor amplitude is modulated by the contact-stiffness and viscous-damping changes associated with lubricant film formation: as the film degrades and asperity contact increases, the ultrasonic level at the bearing housing rises [10]. Commercial instruments (UE Systems, Sonotec) exploit this by comparing the level against a reference baseline and triggering relubrication when a defined threshold is exceeded, making threshold-based passive US the most widely deployed non-invasive LCM method in routine industrial use. The passive US probe used in this study follows the same design: it senses around 38–40 kHz and heterodynes (down-mixes) that band to an audio-frequency baseband, so the recorded US signal is band-limited to 0–20 kHz (Section 3)—a property of the available sensor rather than an inherent limit of ultrasound. However, the level is also sensitive to rotational speed and load independently of lubrication condition, which

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limits the reliability of threshold-based approaches under varying operating conditions.

The two modalities thus respond to the same underlying phenomenon—increasing asperity contact as the lubricant film degrades—but through different physical mechanisms and at different frequency scales: AE captures individual high-frequency transient events (≥ 50 kHz) from discrete micro-fractures and asperity collisions, while passive US, after the probe heterodynes its 38–40 kHz sensing band to a 0–20 kHz baseband, reflects the lower-frequency bulk acoustic envelope comprising structural resonances, friction noise, and the aggregate effect of contact activity. Both channels therefore carry information about κ , and the question of whether they provide complementary or redundant lubrication-relevant information is addressed experimentally in Sections 3–6.

3 Experimental Setup

3.1 Test rig and bearing

The test rig is a custom-built bearing test stand at AU (Herning, Denmark). A single deep-groove ball bearing (model SYJ25, bore 25 mm, pitch diameter $d_{pw} = 38.0$ mm, 9 chrome steel rolling elements, dynamic load rating 14000 N) is driven by a variable-frequency electric motor controlled through a variable-frequency drive (VFD). The bearing is loaded radially by a static weight and lubricated by a defined quantity of Keratech 22 oil (kinematic viscosity $\nu_{40} = 22$ cSt, $\nu_{100} = 4.1$ cSt) .

Note on VFD noise: Harmonic artefacts from the VFD switching frequency are present at approximately 5 kHz and its multiples in the low-frequency range of both sensor signals. These are partially filtered out

3.2 Operating conditions

The experiment follows a protocol which varies the operating conditions through changes in rotational speed (rpm) and housing temperature ($^{\circ}\text{C}$). This consists of a staircase sweep from 100 to 3000 RPM and back down to 100 RPM in steps of 100 RPM with a 60-second hold at each step. This staircase is repeated for each of 13 temperature blocks (40–100 $^{\circ}\text{C}$ in 5 $^{\circ}\text{C}$ increments) where temperature is controlled with . Staircases are separated by a 2-minute standstill (RPM = 0) at the start of each block for thermal stabilisation. The hold at the 3000 rpm staircase peak is substantially longer than the ordinary steps (approximately 20 minutes, while the next temperature set-point is established), so this operating point is strongly over-represented in the data: it accounts for roughly a third of all sweeps. The applied radial load is constant at 2 N throughout the experiment. The total duration of the experiment is approximately 14 hours. The measured operating conditions span 140–2977 rpm and 33–100 $^{\circ}\text{C}$; the measured temperature range extends below the commanded blocks.

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The lubrication conditions in the bearing are not expected to change during the experiment in other ways than what is entailed by rpm and temperature and the effect of these upon the lubricant. It is not expected or assumed that any degradation, starvation or contamination of the lubricant happens.

The operating conditions are controlled through . The measured operating conditions space, distribution and time variation are seen in Figures 2 and 3. The κ value for each sweep is computed according to Equations (1) and (2) using the per-sweep measured RPM and temperature, with ν evaluated via the ASTM D341 Walther viscosity–temperature equation.

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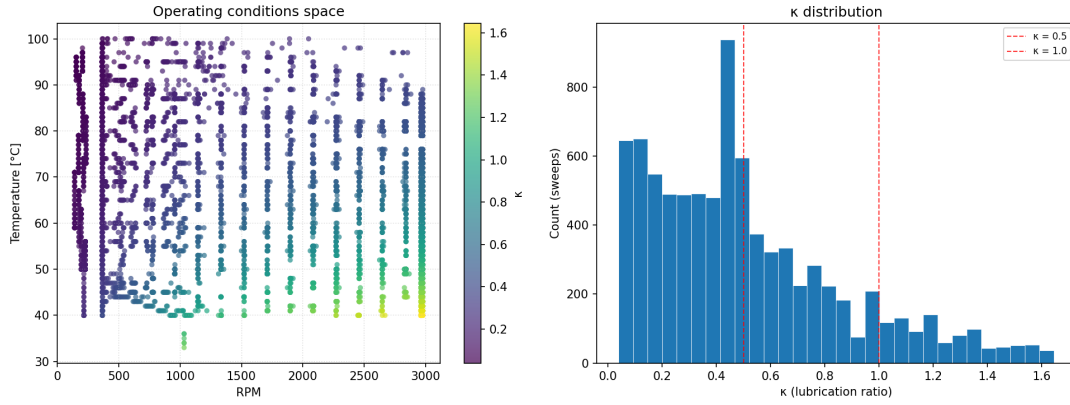


Figure 2: Left: space of the measured operating conditions (RPM and temperature) during the experiment. Each point corresponds to a 20ms window of sampling at a given RPM and temperature. The colour of each point corresponds to the derived κ value for that sweep, with warmer colours indicating higher κ values. Right: distribution fo the derived κ values.

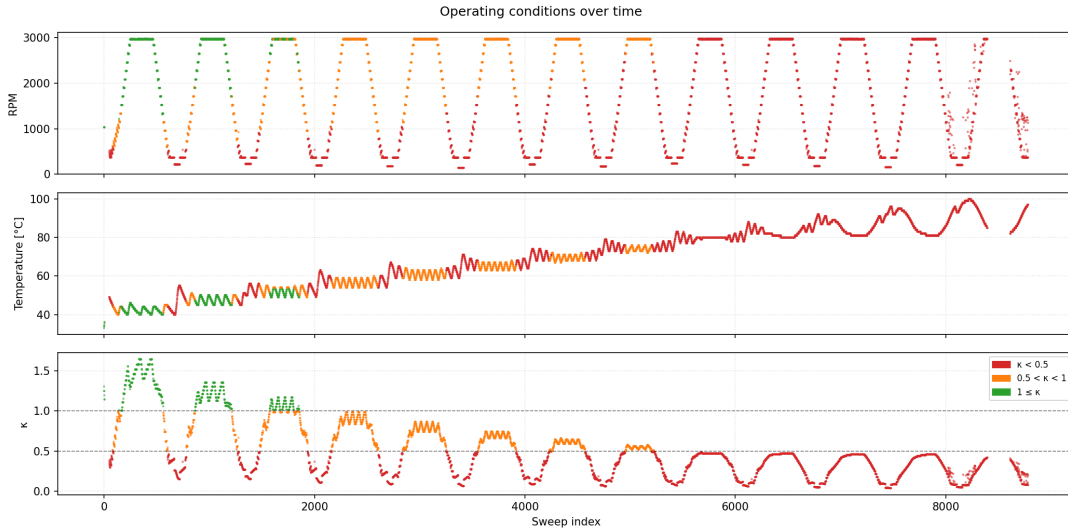


Figure 3: Measured operating conditions (RPM and temperature) for each sweep during the experiment. Each point corresponds to a 20ms window of sampling at a given RPM and temperature. The colour of each point corresponds to the derived κ value for that sweep, with warmer colours indicating higher κ values.

3.3 Sensor configuration and data acquisition

Two sensors simultaneously acquire signals from the bearing housing through a shared oscilloscope which samples at a shared sample rate of 12.5 MHz:

Acoustic emission sensor: A wideband piezoelectric AE sensor is bonded to the bearing housing.

Ultrasound sensor: A heterodyned passive US probe is bonded to the bearing housing.

Temperature sensor:

RPM sensor:

Load sensor:

The shared sample rate and DAQ setup ensures that AE and US signals share timestamps.

Acquisition windows (sweeps). Data are acquired as short windows rather than continuously: every ~ 5.9 s the oscilloscope captures a 20 ms window ($N \approx 2.5 \times 10^5$ samples per channel) containing time-synchronous AE and US waveforms, together with the mean RPM and housing temperature over the window. Throughout this paper, one such window is referred to as a *sweep*, and a sweep is the unit of observation for all feature extraction, model training, and evaluation. After excluding an initial stationary warm-up segment recorded before the first staircase, the protocol yields 8 739 raw sweeps, of which 235 with measured RPM above the 3000 rpm protocol maximum are discarded,

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leaving 8 504 sweeps per channel (median 10 sweeps per staircase hold). The resulting κ distribution spans $[0.04, 1.64]$ with mean 0.51 (Figure 2).

4 Signal Processing and Feature Extraction

4.1 Waveforms and spectrograms

Spectrograms of the raw signals across the operating conditions sorted according to κ values are shown in Figure 4 to illustrate energy of the frequency content and its variation with operating conditions.

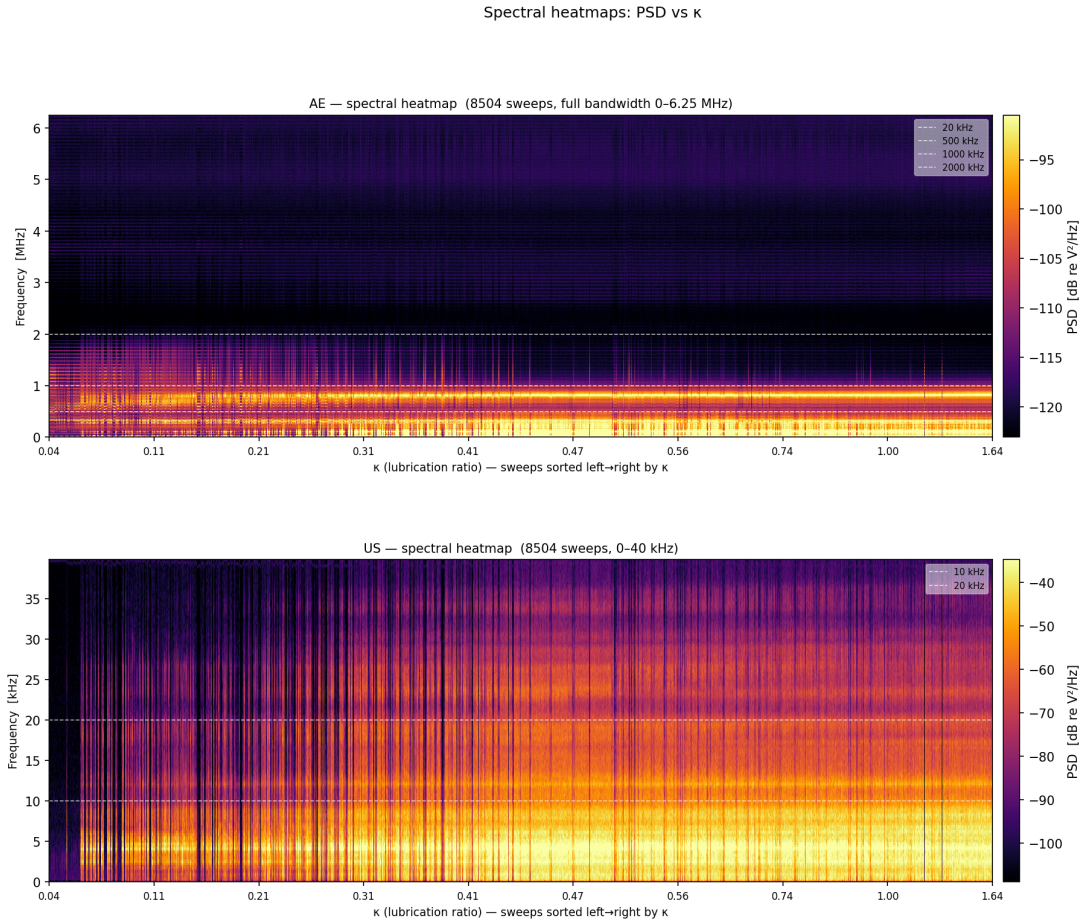


Figure 4: Spectrograms of the raw AE and US signals across the operating conditions. Each column corresponds to a sweep, sorted by increasing κ value from left to right. The colour indicates the magnitude of the signal in dB, with warmer colours indicating higher magnitudes.

4.2 Sub-band decomposition

The AE and US signals are decomposed into frequency sub-bands prior to feature extraction, since the information on lubrication conditions contained in the signals is expected

to be localised in sub-bands rather than wholly contained in the broadband signals [7, 16]. For the AE channel, three sub-bands are extracted in addition to the broadband signal: (i) 20–500 kHz, capturing the transition from audible to ultrasonic contact dynamics; (ii) 500 kHz–1 MHz, the primary stress-wave propagation band; and (iii) 1–2 MHz, covering high-frequency activity associated with film breakdown; the content of this band is validated in Section 4.3 and its physical origin discussed in Section 7.2. The US channel is first restricted to 0–20 kHz, consistent with the heterodyned baseband of the probe (Section 2.2), and two sub-bands (0–10 kHz and 10–20 kHz) are extracted in addition. The sub-bands are summarised in Table 1. All decompositions use 5th-order Butterworth filters applied forward–backward (`sosfiltfilt`), giving zero phase distortion and a 10th-order effective magnitude response.

Table 1: Frequency sub-bands used for signal pre-filtering before feature extraction.

Channel	Sub-band	Filter type	Frequency range
AE	Broadband	–	Full bandwidth
AE	Sub-band 1	Band-pass	20–500 kHz
AE	Sub-band 2	Band-pass	500 kHz–1 MHz
AE	Sub-band 3	Band-pass	1–2 MHz
US	Broadband	Low-pass	0–20 kHz
US	Sub-band 1	Band-pass	0–10 kHz
US	Sub-band 2	Band-pass	10–20 kHz

4.3 Sub-band signal validation

The sub-band interpretation rests on the assumption that the band content is bearing-generated. Because the VFD injects switching EMI—visible as a line comb at ≈ 4.7 kHz spacing and its multiples, matching the VFD switching frequency (Section 3.1)—this assumption is verified rather than presumed. The validation works with *total* band power: spectral redistribution conserves energy within a band, so changes in the comb’s line shape with drive state cannot produce a net change in band power.

Figure 5 shows the total power of each AE band as a function of RPM within two fixed temperature windows, and the three bands behave qualitatively differently. The 20–500 kHz band rises monotonically with speed ($\rho = +0.93$ at 45–55 °C), as expected for mechanically generated contact and rolling noise [7, 12]. The 500 kHz–1 MHz band rises only weakly ($\rho = +0.71$). The 1–2 MHz band is *inversely* related to speed ($\rho = -0.76$): its power falls with speed and reaches a common electronics floor near 783 rpm ($\kappa \approx 0.52$) at 45–55 °C, but persists to 2206 rpm ($\kappa \approx 0.32$) at 85–95 °C, where the film forms later in the speed range. The high-frequency content thus disappears at a comparable lubrication state rather than at a comparable speed: it is rotation-generated and gated by the lubricant film, superimposed on the drive’s EMI comb. The physical mechanism

is discussed in Section 7.2.

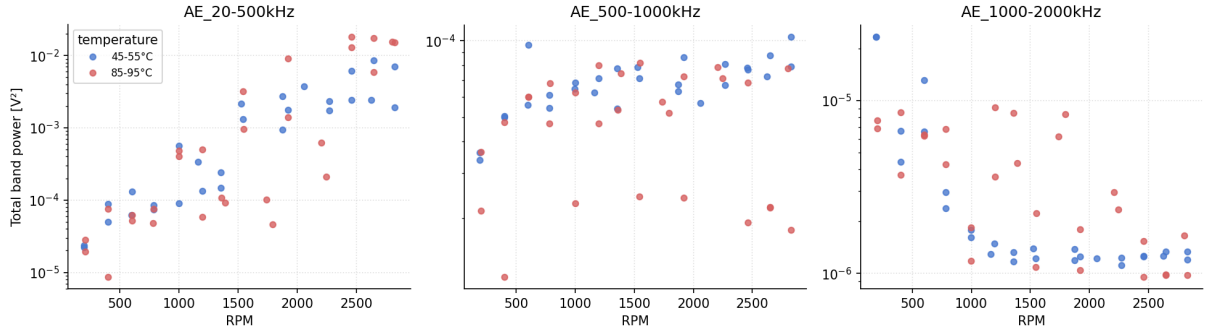


Figure 5: Total AE band power as a function of RPM at two fixed temperature windows. The 20–500 kHz band rises monotonically with speed; the 1–2 MHz band decays to a common electronics floor at a temperature-dependent speed, i.e. at a comparable lubrication state (κ) rather than a comparable RPM.

4.4 Feature extraction

Features are computed from the voltage signal of each sweep–sensor pair, for the broadband signal and for each sub-band. In total 14 features are computed, comprising 8 time-domain and 6 frequency-domain statistics, as listed in Tables 2 and 3. The set is deliberately canonical: commonly used companions such as standard deviation, variance, peak, impulse factor, and RMS frequency are exact transforms or products of the retained features (e.g. $\text{std} = \text{RMS}$ for zero-mean band-filtered signals; $\text{peak} = \text{crest} \times \text{RMS}$; $\text{impulse} = \text{crest} \times \text{shape}$; $f_{\text{RMS}}^2 = f_c^2 + \sigma_w^2$) and are therefore omitted, so that no candidate feature is a deterministic function of another. All retained features (without the specific sub-bands) have previously been applied to signals for bearing condition monitoring and LCM [1, 20]; the Hjorth parameters originate from EEG analysis [21].

All frequency-domain features are computed from the one-sided FFT magnitude spectrum of the sweep. The spectral skewness and kurtosis are standardised by $\sigma_w^p \Sigma S$ and are therefore dimensionless and invariant to overall signal amplitude, so that spectral-shape information is separated from the amplitude information carried by RMS.

4.5 Candidate feature sets

Extracting the 14-feature set independently for the broadband signal and each sub-band yields $14 \times 4 = 56$ candidate AE features and $14 \times 3 = 42$ candidate US features per sweep. Features are named by band and statistic (e.g. `AE_1000-2000kHz__complexity`); broadband features carry no band prefix. Section 5 reduces these candidate sets to compact, non-redundant subsets for modelling.

Table 2: Time-domain features. Notation: N signal length; $\bar{x}$ sample mean; σ population standard deviation; $x_{\text{peak}} = (\max x - \min x)/2$ half the peak-to-peak range.

Name	Formula	Description
RMS	$\sqrt{N^{-1} \sum_n x_n^2}$	Root mean square amplitude.
Skewness	$N^{-1} \sum_n [(x_n - \bar{x})/\sigma]^3$	Third standardised central moment; amplitude asymmetry.
Kurtosis	$N^{-1} \sum_n [(x_n - \bar{x})/\sigma]^4$	Fourth standardised central moment; tail heaviness.
Crest factor	$x_{\text{peak}}/x_{\text{RMS}}$	Peak-to-RMS ratio; sensitive to impulsive events.
Shape factor	$x_{\text{RMS}} / (N^{-1} \sum_n x_n)$	Ratio of RMS to mean absolute value.
Margin factor	$x_{\text{peak}} / (N^{-1} \sum_n \sqrt{ x_n })^2$	Peak divided by squared mean square-root amplitude.
Hjorth mobility	$[\text{Var}(x') / \text{Var}(x)]^{1/2}$	Square root of first-derivative to signal variance ratio; proportional to mean frequency.
Hjorth complexity	$M(x')/M(x)$	Ratio of first-derivative mobility M to signal mobility; bandwidth proxy.

Table 3: Frequency-domain features (companion to Table 2). Notation: $S_k = |X_k|$ one-sided FFT magnitude at bin k ; f_k corresponding frequency; $\bar{S}$ arithmetic mean of $\{S_k\}$; $\Sigma S = \sum_k S_k$; $f_c = (\Sigma S)^{-1} \sum_k f_k S_k$; $\sigma_w = ((\Sigma S)^{-1} \sum_k (f_k - f_c)^2 S_k)^{1/2}$.

Name	Formula	Description
Dominant frequency	$f_{\arg \max_k S_k^2}$	Frequency at which spectral power is maximum.
Centre frequency	$f_c = \sum_k f_k S_k / \Sigma S$	Magnitude-weighted mean frequency (spectral centroid).
Spectral width	$\sigma_w = (\sum_k (f_k - f_c)^2 S_k / \Sigma S)^{1/2}$	Magnitude-weighted spectral spread about the centroid.
Spectral skewness	$\sum_k (f_k - f_c)^3 S_k / (\sigma_w^3 \Sigma S)$	Standardised third weighted spectral moment; asymmetry of the spectral distribution.
Spectral kurtosis	$\sum_k (f_k - f_c)^4 S_k / (\sigma_w^4 \Sigma S)$	Standardised fourth weighted spectral moment; peakedness/tail weight of the spectral distribution.
Spectral flatness	$\exp(K'^{-1} \sum_{k:S_k > 0} \ln S_k) / \bar{S}$	Geometric-to-arithmetic mean ratio; 1 for white noise, 0 for a pure tone.

5 Feature Analysis

This section reduces the 56 AE and 42 US candidate features (Section 4) to compact, non-redundant subsets for modelling, and characterises what the surviving features respond to. The analysis proceeds in two selection stages – a relevance filter against κ followed by a redundancy reduction – after which the operating-condition dependence and low-dimensional structure of the retained sets are examined.

5.1 Two-stage feature selection

Stage 1 – relevance. Each candidate feature is ranked by its absolute Spearman ($|\rho|$, monotonic association) and Pearson ($|r|$, linear association) correlation with κ across all sweeps. Features must satisfy both $|\rho| \geq 0.1$ and $|r| \geq 0.1$ to be retained, removing features carrying essentially no monotonic or linear information about the target.

Stage 2 – redundancy. Among the survivors, redundancy is reduced by a greedy pairwise procedure. Features are visited in order of increasing relevance ($|\rho|$ with κ); whenever two features are inter-correlated at $|\rho| \geq 0.90$ and the candidate for removal also has a variance inflation factor $VIF \geq 10$, the member of the pair with the higher relevance is kept and the other is dropped. The VIF thus acts as a qualifier for pairwise removal rather than as a stand-alone criterion: no feature is removed for high VIF alone, and a retained feature may still exhibit $VIF > 10$ when its collinearity is distributed across several features rather than concentrated in a single partner.

Table 4 summarises the selection funnel.

Table 4: Feature selection funnel: candidate features, survivors of the Stage 1 relevance filter, and the final retained sets after Stage 2 redundancy reduction.

Channel	Candidates	After Stage 1	Retained
AE	56	49	27
US	42	26	10
Combined	–	–	37

5.2 Correlation with κ

Table 5 lists the top ten features for each sensor ranked by combined Spearman–Pearson rank score with κ ; the complete rankings of all candidates, including Stage 2 retention, are given in Appendix A (Tables 9 and 10).

Table 5: Top 10 features by combined Spearman–Pearson rank score with κ , for the AE and the US signals. Features retained after Stage 2 redundancy reduction are marked with † .

AE			US		
Feature	$ \rho $	$ r $	Feature	$ \rho $	$ r $
1–2 MHz complexity †	0.892	0.736	10–20 kHz RMS †	0.691	0.609
500 kHz–1 MHz mobility	0.848	0.662	RMS	0.614	0.548
500 kHz–1 MHz centre freq. †	0.799	0.687	0–10 kHz RMS	0.610	0.542
1–2 MHz shape factor	0.827	0.634	10–20 kHz mobility †	0.583	0.536
1–2 MHz RMS	0.781	0.551	0–10 kHz kurtosis	0.465	0.331
Shape factor †	0.704	0.621	Kurtosis †	0.468	0.319
1–2 MHz spectral bandwidth †	0.683	0.584	10–20 kHz crest factor †	0.350	0.394
500 kHz–1 MHz dominant freq. †	0.676	0.582	0–10 kHz mobility †	0.380	0.348
500 kHz–1 MHz complexity †	0.682	0.529	10–20 kHz complexity †	0.342	0.306
20–500 kHz shape factor	0.631	0.554	10–20 kHz margin factor	0.280	0.358

The top-ranked AE feature is 1–2 MHz complexity ($|\rho| = 0.892$), and the top-ranked US feature is 10–20 kHz RMS ($|\rho| = 0.691$). Broadband AE features rank lower than their sub-band counterparts, indicating that the frequency-specific content of the high-frequency AE signal carries more monotonic information about κ than the full-band aggregate [7, 16].

5.3 Redundancy structure

Figure 6 shows the variance inflation factors of the Stage 1 survivors on a logarithmic axis. Highly inter-correlated pairs lose their less κ -relevant member in Stage 2 (Section 5.1); retained features above the $VIF = 10$ line are those whose collinearity is distributed across several features rather than concentrated in a single pairwise partner.

Interpret the final ranking: which sub-bands dominate for each channel; amplitude (RMS) vs. shape (complexity, bandwidth, skewness) features;

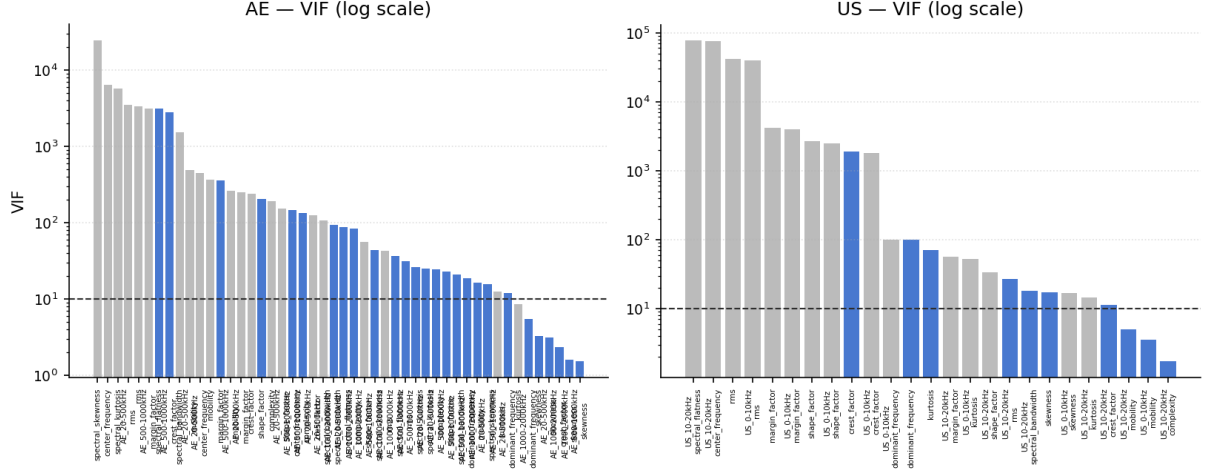


Figure 6: Variance inflation factors of the Stage 1 survivors (logarithmic axis). Retained features in blue, removed features in grey; the dashed line marks the $VIF = 10$ qualifier used by the Stage 2 pairwise removal. Values are capped at 10^6 for display.

5.4 Operating-condition dependence

The raw correlation between signal features and κ is partially driven by the shared dependence on rotational speed and temperature: RPM has a Spearman correlation of $\rho = ??$ with κ and temperature has $\rho = ??$. Higher RPM reduces the required viscosity ν_1 (Eq. (2)) and therefore increases κ , while simultaneously increasing signal energy through more frequent contact events per second. Higher temperature reduces the actual viscosity ν and therefore reduces κ , while also affecting material damping and structural dynamics.

As a consequence, amplitude features such as RMS are partially acting as indirect proxies for rotational speed rather than as direct indicators of lubrication film adequacy [7, 12, 15]. This confounding is inherent to any single-operating-point sensor system without an explicit operating-condition normalisation step. Because κ is fully determined by the operating point (Section 2.1), a model given measured RPM and temperature directly would predict κ near-perfectly by construction; the contribution of the acoustic features is to recover this information without tachometer or thermometer access. This proxy structure is quantified directly. A LightGBM regressor trained on the retained AE features predicts RPM with $R^2 = 0.965$ (RMSE 206 rpm) and temperature with $R^2 = 0.915$ (RMSE 4.6 °C) on the hold-out set. Feeding these predicted operating points through Eqs. (1)–(2) yields a two-stage κ estimate with $R^2 = 0.950$ (RMSE 0.081), nearly matching the direct features-to- κ model ($R^2 = 0.953$, RMSE 0.079). The direct model is therefore well described as an acoustic operating-point soft sensor combined with the ISO 281 mapping, with only a marginal contribution beyond it.

The features are not, however, mere speed trackers. Within a fixed staircase RPM step the drive state is approximately constant while the lubricant viscosity varies across the 13 temperature blocks; in this setting the strongest feature (1–2 MHz complexity) retains a

median within-step Spearman correlation of -0.68 with temperature (negative in 91% of steps, i.e. tracking κ ; Figure 7), while remaining constant during the stationary heater-driven warm-up. The features therefore sense both κ channels—speed and temperature—through the rotating bearing, rather than reading either operating quantity directly.

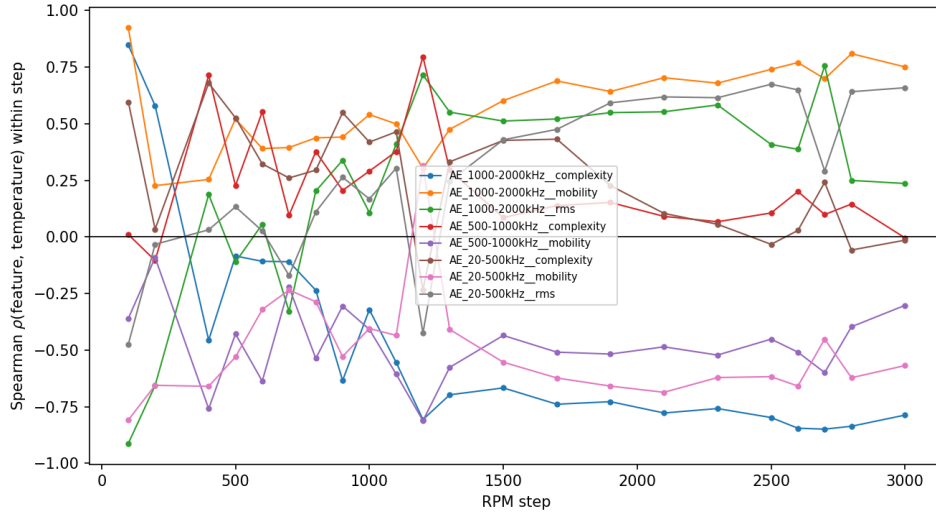


Figure 7: Within-RPM-step Spearman correlation between selected AE sub-band features and temperature. At fixed speed the drive state is constant, so non-zero correlations reflect sensitivity to the lubrication state rather than to the operating point.

5.5 Low-dimensional structure and regime separability

Figure 8 shows the first two principal components of the retained AE and US feature sets coloured by κ value. Visual inspection shows that some separability in this subspace is possible, confirming that the retained feature sets carry regime-relevant discriminative information.

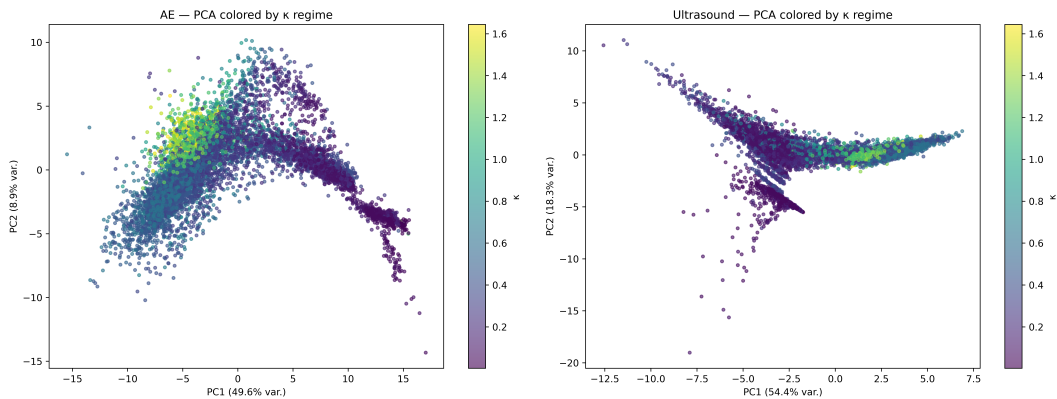


Figure 8: PCA projection of the retained AE and US feature sets, coloured by lubrication κ value.

6 Regression Modelling and Statistical Evaluation

6.1 Experimental design

6.1.1 Data sets and splits

The data experiments utilise three different features sets: the AE, the US, and the combined AE and US feature sets. Each of the later described models are fitted to each of the three datasets. Each dataset is divided into a training set (80%) and a hold-out test set (20%) by sweep-level random sampling, stratified to preserve the κ regime distribution. The hold-out set is withheld entirely from all model development steps and used only for final evaluation. The training set is further divided during nested cross-validation described in Section 6.2.

6.1.2 Model families

Three regression model families are evaluated for three sensor configurations, as summarised in Table 6.

Model	Rationale	Hyperparameters & tuning
Elastic Net	$\ell_1 + \ell_2$ regularised linear model; ℓ_1 promotes sparsity, ℓ_2 stabilises correlated features.	α (strength) and ℓ_1 ratio; ElasticNetCV inner CV loop.
Polynomial (deg. 2)	Augments linear model with all pairwise interaction terms; remains linear in parameters.	Ridge penalty λ ; RidgeCV within each outer fold.
LightGBM	Gradient-boosted trees; captures non-linear feature- κ mappings and high-order interactions.	Learning rate, num. leaves, max. depth, min. child samples, ℓ_2 reg.; Bayesian optimisation (Optuna TPE, 20 trials per outer fold).

Table 6: Regression model families: rationale and hyperparameter tuning strategy.

6.2 Repeated nested cross-validation

To obtain stable performance estimates that properly account for the variance introduced by the train/validation split, the model evaluation uses a repeated nested k -fold cross-validation scheme. The outer loop is repeated $R = 5$ times with independently re-seeded 5-fold splits, yielding $R \times k = 25$ outer RMSE scores per model-sensor combination. See Figure 9 for data partitions.

Add Elastic-NetCV search grid values (number of α candidates and ℓ_1 ratios tried)

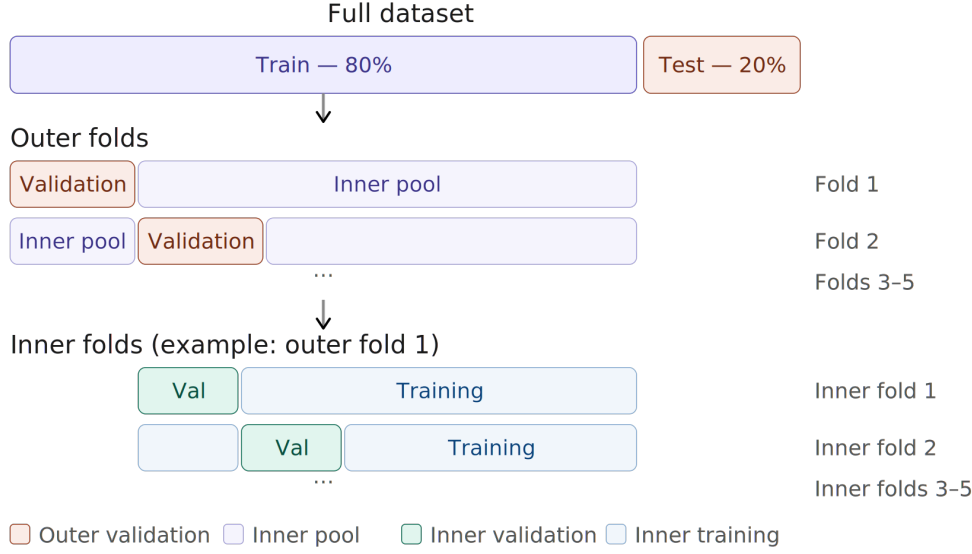


Figure 9: Caption

For LightGBM, each outer fold contains its own inner Optuna optimisation loop, ensuring that the reported outer scores are not contaminated by hyperparameter selection on the outer fold data. For the linear models, hyperparameter selection is handled by ElasticNetCV and RidgeCV within each outer training partition.

The 25 outer scores form the empirical sampling distribution of RMSE for each model and are used directly in the significance tests described below.

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6.3 Statistical significance testing

Pairwise significance tests are conducted at two levels.

Level 1 — Within-sensor comparisons (architecture differences). For pairs of models trained on the same sensor configuration (e.g., Elastic Net vs. Polynomial on AE), the corrected repeated k -fold t -test of Nadeau and Bengio [22] is applied to the 25 paired RMSE scores. This test corrects the standard paired t -test for the positive correlation induced by shared training data across repeated folds. Multiple within-sensor comparisons are adjusted by the Holm–Bonferroni step-down procedure at $\alpha = 0.05$.

Level 2 — Cross-sensor comparisons (best per sensor configuration). For the best model from each sensor configuration (selected by lowest mean CV RMSE), pairwise differences on the common hold-out set are assessed by:

- The Wilcoxon signed-rank test on paired residuals (non-parametric);
- The modified Diebold–Mariano test for equal predictive accuracy under squared-error loss;

- A bootstrap Δ RMSE 95% confidence interval (10 000 resamples with replacement).

Cross-sensor p -values are also corrected by Holm–Bonferroni.

6.4 Results

Model	Sensors	CV RMSE	HO R^2	HO MAE	HO RMSE
LightGBM	Combined	0.0722 ± 0.0037	0.962	0.045	0.071
LightGBM	AE	0.0798 ± 0.0036	0.953	0.052	0.079
Elastic Net	Combined	0.1518 ± 0.0048	0.833	0.113	0.149
LightGBM	US	0.1603 ± 0.0064	0.815	0.102	0.156
Elastic Net	AE	0.1621 ± 0.0050	0.808	0.123	0.159
Polynomial	Combined	0.1636 ± 0.0079	0.780	0.131	0.171
Polynomial	AE	0.1693 ± 0.0037	0.797	0.124	0.164
Polynomial	US	0.1931 ± 0.0052	0.723	0.134	0.192
Elastic Net	US	0.2278 ± 0.0049	0.596	0.171	0.231

Table 7: Regression model performance. CV RMSE: mean $\pm$ std of outer RMSE across 25 repeated nested CV folds (training set $n = 6\,794$). HO: hold-out test set ($n = 1\,710$). Best hold-out R^2 in bold.

Overall performance. LightGBM achieves the best hold-out performance across all sensor configurations, with AE-only reaching $R^2 = 0.953$, RMSE = 0.079 κ -units. On a target range of $[0.04, 1.64]$ with mean 0.51, this corresponds to a relative RMSE of approximately ??% of the mean target value. Linear and polynomial models on AE features achieve $R^2 = 0.797$ – 0.808 , confirming a non-linear feature– κ mapping that gradient boosting exploits more effectively.

Sensor comparison. AE features substantially outperform US features across all model families. The LightGBM AE–US gap ($\Delta R^2 = 0.14$) is confirmed as highly significant (Nadeau–Bengio $p = 2.3 \times 10^{-21}$; Wilcoxon $p = 8.8 \times 10^{-74}$; Δ RMSE = 0.0776, 95% CI $[0.0671, 0.0880]$). This reflects the richer frequency content and more direct physical sensitivity of the AE sensor to asperity contact events.

Sensor fusion. LightGBM on the combined AE+US feature set achieves $R^2 = 0.962$ ($\Delta R^2 = 0.009$ over the AE-only result of 0.953). The within-CV corrected t -test gives $p = 1.5 \times 10^{-5}$; on the hold-out set, Wilcoxon $p = 5.5 \times 10^{-16}$, Diebold–Mariano $p = 1.6 \times 10^{-8}$, and the bootstrap Δ RMSE = 0.0079 with 95% CI $[0.0053, 0.0105]$. For the linear models, the combined feature set performs similarly to AE alone (Elastic Net: R^2 0.833 vs. 0.808; Polynomial: 0.780 vs. 0.797). The fusion gain is small in absolute terms but consistent across all four tests, indicating that the US channel contributes a modest amount of non-redundant information.

Model consistency. The close agreement between nested CV RMSE and hold-out RMSE across all models indicates stable generalisation with no evidence of significant overfitting.

Figure 10 shows predicted vs. actual κ for the hold-out set for all model families under the AE-only configuration.

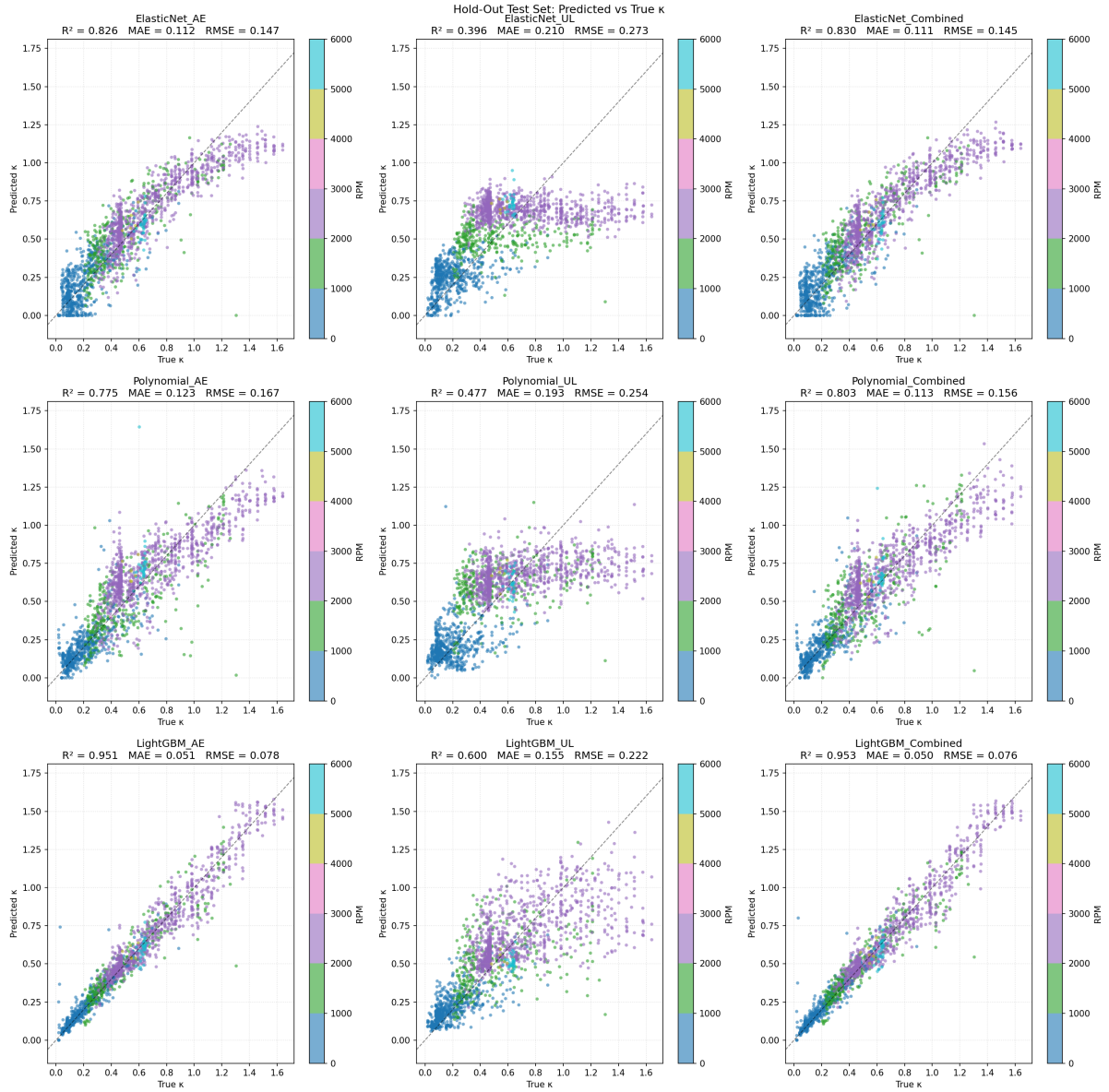


Figure 10: Predicted vs. actual κ on the hold-out test set for all model families and sensor configurations, colored according to RPM. The dashed diagonal indicates perfect prediction. Residual spread increases in the mixed lubrication regime ($0.5 \leq \kappa < 1.0$).

SHAP-based feature importance. SHAP (SHapley Additive exPlanations) values computed via TreeExplainer identify which AE sub-band features drive the κ predictions. The dominant contributor by a large margin is *1–2 MHz complexity* (mean $|\text{SHAP}| = 0.168$), followed by *500 kHz–1 MHz centre frequency* (0.078) and *1–2 MHz mobility* (0.044). The 1–2 MHz complexity feature ranks *first* for all three model types, and the 500 kHz–1 MHz centre frequency also appears in all three top-10s (cross-model agreement, Table 8), making 1–2 MHz complexity the most robust single predictor of κ in the dataset. [Figure 11](#)

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Feature	ElasticNet	Polynomial	LightGBM	Models in top 10
1–2 MHz complexity	1	1	1	3
500k–1 MHz centre freq.	7	8	2	3
Broadband spectral flatness	2	—	5	2
1–2 MHz mobility	3	—	3	2
500k–1 MHz RMS	5	—	9	2
1–2 MHz dominant freq.	6	—	8	2
500k–1 MHz dominant freq.	—	5	6	2

Table 8: Cross-model SHAP agreement for AE features: features appearing in the top 10 by mean $|\text{SHAP}|$ for more than one model type.

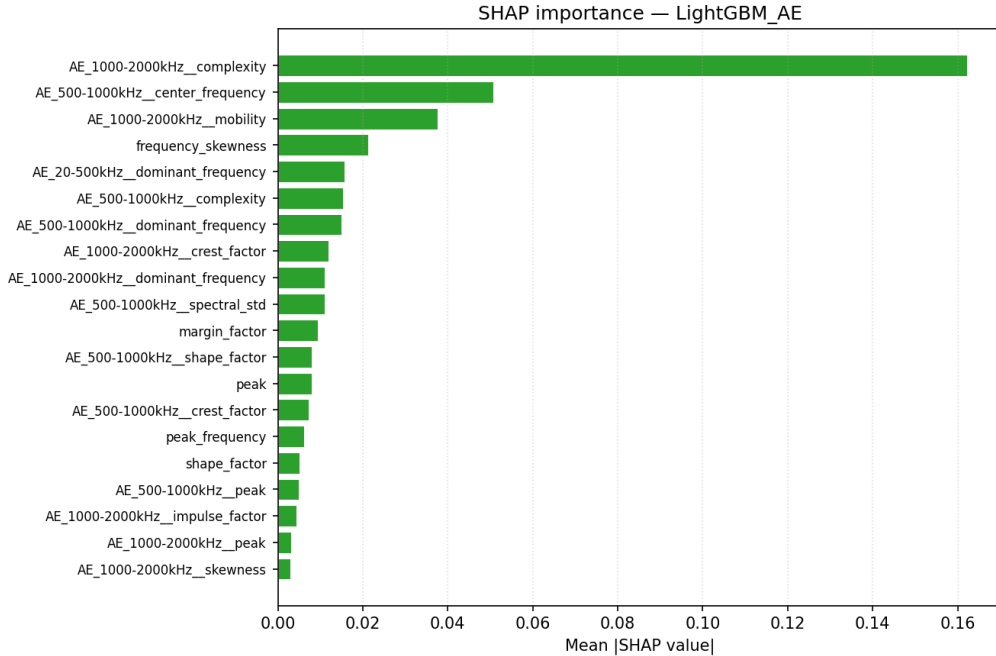


Figure 11: SHAP feature importance (mean $|\text{SHAP value}|$) for the LightGBM model trained on AE features. Features from frequency sub-bands are prefixed with the band label (e.g., AE_20–500kHz_).

7 Discussion

7.1 AE vs. US

The consistently superior performance of AE over the US channel across all model families reflects a fundamental hardware difference. The AE sensor captures wideband, rotation-generated content extending into the MHz range (analysed here up to 2 MHz): verified stress-wave emission up to several hundred kHz, and film-gated high-frequency activity above (Sections 4.3 and 7.2). The US channel is heterodyned, limiting its recorded content

to a 0–20 kHz baseband in which the signal is a mixture of down-mixed contact activity, low-frequency machinery noise, VFD harmonics, and structural vibration; within this baseband, the 10–20 kHz half carries the strongest κ correlations (Section 5.2). The physical information about microscale contact events is largely absent in this frequency range. Replacing the heterodyned probe with a wideband passive acoustic or ultrasound sensor would be expected to substantially improve the US-channel models.

7.2 Origin of the high-frequency AE content

The validation of Section 4.3 shows that the 1–2 MHz band superimposes a VFD switching comb on a broadband, rotation-generated component that vanishes once the lubricant film forms. Two mechanisms are compatible with this behaviour. The first is mechanical: asperity-contact emission whose high-frequency tail remains detectable despite the strong structure-borne attenuation at MHz frequencies. The second is electrical: VFD common-mode voltage discharging through the bearing while the film is thin (bearing currents), producing broadband high-frequency transients that are directly film-gated and would also explain the prominence of drive-related structure in the same band. Both mechanisms are film-mediated, so the band’s predictive value for κ is genuine in either case; distinguishing them—for instance via shaft-voltage measurement or an electrically insulated sensor mount—is left to future work. For deployment, the electrical hypothesis implies that the high-frequency features may be drive-specific, whereas the verified 20–500 kHz band, whose energy rises monotonically with speed, is expected to transfer across drive configurations.

Add bearing-current/ED references.

7.3 Fusion gain

The fusion gain ($\Delta R^2 = 0.009$, $\Delta \text{RMSE} = 0.0079$ on the hold-out) is small but statistically robust, detected under repeated nested cross-validation and confirmed by all three hold-out tests. Its limited magnitude can be understood from two complementary angles. First, the US channel retains only 10 features after selection (vs. 27 for AE), and the top US correlations ($|\rho| \leq 0.691$) are weaker than the top AE correlations ($|\rho| = 0.892$). The US channel therefore contributes a genuine but limited amount of non-redundant information to the combined model. Second, the US channel’s heterodyned 0–20 kHz baseband means its information content partially overlaps with the low-frequency tail of the AE signal, limiting the independent contribution it can make.

This result is an informative contribution to the LCM literature, where AE–ultrasound complementarity is widely *argued* but rarely *tested*. Whether a wideband passive ultrasound sensor (rather than a heterodyned probe) would provide more independent information – and thus a more substantial fusion gain – remains an open question for future work.

7.4 Interpreting the amplitude-confounding result

The strong raw Spearman correlations of amplitude features (RMS and related statistics) and frequency-tracking features (Hjorth mobility, centre frequency) with κ should be interpreted with caution. As discussed in Section 5.4, rotational speed is a driver of both κ (through ν_1 in Eq. (2)) and broadband signal energy (through the rate of and energy in contact events per revolution as described in [7]). A model trained on amplitude features therefore exploits speed as an indirect proxy for κ rather than sensing purely lubrication-specific signal changes directly.

This confounding does not invalidate the regression results: a model that uses speed-mediated amplitude changes to predict κ is still predicting κ , and its predictions may be operationally useful. However, it does indicate that the model’s generalisation to operating conditions outside its training distribution (different speed ranges, different bearing loads) may be limited. The two-stage analysis of Section 5.4 makes this concrete: predicted operating points alone, fed through the ISO 281 mapping, recover $R^2 = 0.950$ of the direct model’s $R^2 = 0.953$ —the model is, to first order, an operating-point soft sensor, and its lubrication sensitivity beyond the operating point rests on the within-step evidence of Section 5.4. Future feature engineering should prioritise amplitude-invariant spectral-shape features (spectral skewness and kurtosis, spectral bandwidth, spectral flatness) that are expected to carry a higher proportion of genuine lubrication-state information relative to operating condition proxies.

7.5 Implications for the κ regression framework

The κ regression framework is validated as a methodologically sound approach for passive non-invasive LCM, when κ is assumed to be representative of the lubrication conditions in the bearing. A hold-out $R^2 = 0.953$ and $\text{RMSE} = 0.079$ κ -units for a single AE sensor represents strong performance across a demanding continuous speed-and-temperature ramp spanning boundary through full-film lubrication conditions. This exceeds the performance reported by Jakobsen et al. [1] in a comparable passive sensing study with US.

The repeated nested cross-validation design provides appropriately conservative performance estimates: the 25 outer RMSE scores capture fold-to-fold variance without the optimistic bias introduced by simple k -fold evaluation with shared hyperparameter search. The corrected significance tests confirm that performance differences between model families are assessed under assumptions that properly account for data re-use.

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7.6 Limitations

1. **Single bearing and lubricant.** All results are specific to the SYJ25 bearing and Keratech 22 oil. Generalisability to other bearing types, sizes, or lubricants is unknown.
2. **Heterodyned US channel.** The passive US channel is frequency-limited to ≈ 20 kHz. It should not be equated with wideband ultrasound and results should be interpreted in light of this hardware constraint.
3. **Data gap.** _____
4. **Model-derived κ reference.** _____

8 Conclusion

This paper has presented a controlled study of simultaneous acoustic emission and passive ultrasound sensing regressed to the ISO 281 viscosity ratio κ for non-invasive lubrication condition monitoring of a ball bearing. The main findings are as follows:

1. **AE sub-band features are strong predictors of κ .** LightGBM on 27 retained AE features achieves hold-out $R^2 = 0.953$, RMSE = 0.079 κ -units. The 1–2 MHz complexity feature is the dominant SHAP contributor, ranked first for all three model types, identifying the high-frequency AE sub-band as the most informative frequency range for lubrication regime prediction.
2. **Sensor fusion yields a small but consistently significant gain.** The LightGBM combined model achieves $R^2 = 0.962$ ($\Delta R^2 = 0.009$ over AE alone); within-CV corrected t -test $p = 1.5 \times 10^{-5}$; hold-out tests: Wilcoxon $p = 5.5 \times 10^{-16}$, Diebold–Mariano $p = 1.6 \times 10^{-8}$, bootstrap 95% CI [0.0053, 0.0105] excluding zero. The heterodyned US channel thus supplies a modest amount of non-redundant information, though its 10 retained features are individually weaker than the AE features.
3. **AE strongly and significantly outperforms the US channel.** The AE–US gap ($\Delta R^2 = 0.14$ for LightGBM) is confirmed highly significant by all test statistics, reflecting the fundamental bandwidth limitation of the heterodyned US probe.
4. **Rigorous evaluation via repeated nested cross-validation.** The 25 outer RMSE scores per model provide stable, conservative performance estimates; corrected significance tests confirm that all pairwise model differences within each sensor configuration are significant after Holm–Bonferroni correction, with LightGBM strongest throughout.

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5. **The κ regression framework is validated** as a highly effective approach for passive non-invasive LCM, achieving substantially higher performance than prior comparable studies, attributed to the sub-band feature set and the large continuous-sweep dataset.

Future work should address the identified limitations by: acquiring wideband passive ultrasound to replace the heterodyned probe; validating across multiple independent sessions, bearing types, and lubricants; investigating temporal autocorrelation and its effect on hold-out validity; and developing physically motivated sub-band ratio features that are explicitly designed to be robust to operating-condition variation.

A Full Feature Correlation Rankings

Tables 9 and 10 list the complete Spearman and Pearson correlations with κ for all AE and US features that passed the initial data quality filter, sorted by combined rank. Features retained after the Stage 2 VIF and pairwise-correlation redundancy reduction are indicated with a checkmark.

Table 9: Full AE feature correlation ranking with κ . $|\rho|$: absolute Spearman correlation; $|r|$: absolute Pearson correlation. Retained: selected after Stage 2 redundancy reduction.

Rank	Sub-band	Feature	$ \rho $	$ r $	Ret.
1	1000–2000kHz	complexity	0.892	0.736	✓
2	500–1000kHz	mobility	0.848	0.662	–
3	500–1000kHz	center frequency	0.799	0.687	✓
4	1000–2000kHz	shape factor	0.827	0.634	–
5	1000–2000kHz	rms	0.781	0.551	–
6	Broadband	shape factor	0.704	0.621	✓
7	1000–2000kHz	spectral bandwidth	0.683	0.584	✓
8	500–1000kHz	dominant frequency	0.676	0.582	✓
9	500–1000kHz	complexity	0.682	0.529	✓
10	20—500 kHz	shape factor	0.631	0.554	–
11	Broadband	spectral flatness	0.623	0.553	✓
12	Broadband	center frequency	0.613	0.545	–
13	Broadband	mobility	0.584	0.573	–
14	500–1000kHz	rms	0.617	0.528	✓
15	20—500 kHz	center frequency	0.563	0.543	–
16	20—500 kHz	mobility	0.532	0.550	–
17	500–1000kHz	crest factor	0.553	0.523	✓
18	Broadband	margin factor	0.591	0.477	✓

(continued)

Rank	Sub-band	Feature	$ \rho $	$ r $	Ret.
19	Broadband	spectral skewness	0.613	0.446	—
20	1000–2000kHz	crest factor	0.617	0.429	✓
21	500–1000kHz	margin factor	0.509	0.512	—
22	20—500 kHz	complexity	0.523	0.473	✓
23	Broadband	spectral bandwidth	0.575	0.406	—
24	Broadband	spectral kurtosis	0.608	0.362	—
25	Broadband	complexity	0.568	0.380	—
26	20—500 kHz	rms	0.603	0.339	—
27	Broadband	rms	0.602	0.319	—
28	1000–2000kHz	skewness	0.570	0.334	✓
29	20—500 kHz	dominant frequency	0.381	0.521	✓
30	1000–2000kHz	dominant frequency	0.345	0.538	✓
31	500–1000kHz	kurtosis	0.469	0.404	—
32	20—500 kHz	margin factor	0.464	0.343	—
33	Broadband	crest factor	0.505	0.266	—
34	1000–2000kHz	spectral flatness	0.430	0.333	✓
35	Broadband	kurtosis	0.504	0.203	—
36	20—500 kHz	spectral kurtosis	0.390	0.267	✓
37	20—500 kHz	spectral skewness	0.372	0.220	✓
38	500–1000kHz	shape factor	0.296	0.289	✓
39	1000–2000kHz	mobility	0.342	0.222	✓
40	500–1000kHz	skewness	0.364	0.198	✓
41	500–1000kHz	spectral kurtosis	0.298	0.258	✓
42	20—500 kHz	crest factor	0.371	0.157	—
43	20—500 kHz	skewness	0.209	0.261	✓
44	Broadband	skewness	0.181	0.250	✓
45	20—500 kHz	spectral flatness	0.246	0.165	✓
46	20—500 kHz	spectral bandwidth	0.241	0.164	—
47	500–1000kHz	spectral skewness	0.198	0.157	✓
48	500–1000kHz	spectral bandwidth	0.135	0.105	✓

Table 10: Full US feature correlation ranking with κ . $|\rho|$: absolute Spearman correlation; $|r|$: absolute Pearson correlation. Retained: selected after Stage 2 redundancy reduction.

Rank	Sub-band	Feature	$ \rho $	$ r $	Ret.
1	10–20kHz	rms	0.691	0.609	✓
2	Broadband	rms	0.614	0.548	–
3	0–10kHz	rms	0.610	0.542	–
4	10–20kHz	mobility	0.583	0.536	✓
5	0–10kHz	kurtosis	0.465	0.331	–
6	Broadband	kurtosis	0.468	0.319	✓
7	10–20kHz	crest factor	0.350	0.394	✓
8	0–10kHz	mobility	0.380	0.348	✓
9	10–20kHz	complexity	0.342	0.306	✓
10	10–20kHz	margin factor	0.280	0.358	–
11	Broadband	dominant frequency	0.322	0.306	✓
12	0–10kHz	dominant frequency	0.322	0.304	–
13	10–20kHz	kurtosis	0.232	0.347	–
14	Broadband	margin factor	0.276	0.298	–
15	Broadband	crest factor	0.286	0.270	✓
16	0–10kHz	margin factor	0.260	0.289	–
17	0–10kHz	crest factor	0.269	0.253	–
18	Broadband	shape factor	0.236	0.264	–
19	Broadband	skewness	0.289	0.202	✓
20	0–10kHz	shape factor	0.225	0.254	–
21	0–10kHz	skewness	0.276	0.190	–
22	10–20kHz	shape factor	0.113	0.305	–
23	10–20kHz	spectral bandwidth	0.167	0.136	✓
24	10–20kHz	spectral flatness	0.164	0.112	–
25	10–20kHz	center frequency	0.158	0.108	–

CRediT authorship contribution statement

Frederik Appel Vardinghus-Nielsen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interests

The author declares that there are no competing interests.

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IFD?

Data availability

The complete dataset (HDF5 signal files, extracted features in Parquet format, and the reproducible Python analysis pipeline) will be deposited on Zenodo upon acceptance [DOI to be assigned].

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted writing tools for language editing and structural drafting. After using these tools, the author reviewed and edited the content as necessary and takes full responsibility for the content of the publication.

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